

Multimodal AI Product Pipeline

Build something where the AI capability is the product, not a feature bolted on.

OVERVIEW

Most AI-integrated products treat the AI as an add-on: you have a product, and somewhere in it there is a button that calls an LLM. The result is usually underwhelming because the product was not designed around the AI capability from the start.

This assignment asks you to build a small but complete product where a multimodal AI capability is the core of the user loop, not an optional feature. The product must accept at least two modalities as input (image plus text, audio plus text, document plus image) and produce a structured output that is genuinely useful and non-trivial to produce without AI.

We are not assessing whether you can call a multimodal API. We are assessing whether you can design a product experience around an AI capability that is actually worth using.

Why this matters at Heva AI

Several products in our portfolio are built on multimodal AI. We need engineers who can think about product design and AI capability together, not sequentially.

WHAT YOU ARE BUILDING

Define a specific user problem that requires combining at least two input modalities. Build a complete product that solves it. The product must have a real user flow with at least 3 meaningful screens or states, real persistence, and output that is structured and exportable.

Example directions (you are not limited to these): a contract review tool that takes a scanned document and answers specific questions about it in structured form; a meeting intelligence tool that takes an audio recording and an agenda image and produces structured action items mapped to agenda items; a product feedback tool that takes screenshots and voice notes and produces a structured feature request with priority signals. The problem must be yours. Do not implement one of these examples directly.

REQUIREMENTS

Multimodal Input

- Accept at least two distinct input modalities. At least one must be non-text (image, audio, PDF, video).
- Inputs must be processed together, not independently. The AI must reason across both modalities in a single inference pass or a tightly coupled multi-pass pipeline.
- Show the user what was understood from each modality before producing the final output. This is not optional: it is how you build trust in a multimodal product.

AI Pipeline

- The AI pipeline must have at least 2 distinct processing steps. Each step must have a defined input schema and output schema.

- Implement structured output extraction: the final output must be a typed data structure, not free-form text.
- Implement a confidence or uncertainty signal on the structured output. Fields the model is uncertain about must be flagged, not silently included.
- The pipeline must be observable: the user can see which step is running and what it produced.

Product Completeness

- At least 3 meaningful product states: input collection, processing with progress, and structured output review.
- The user can edit the structured output after it is generated. Edits are tracked.
- All outputs are persisted and retrievable. The user has a history of past runs.
- Export: the structured output can be exported as JSON and as a formatted PDF or markdown document.

Performance

- The processing pipeline must show progressive output: as each step completes, partial results appear. The user must not stare at a blank screen waiting for everything to finish.
- Implement a processing time estimate shown to the user when the pipeline starts.

WHAT WE WILL EVALUATE

- Whether the multimodal inputs are actually processed together or just concatenated and sent to the model separately
- The quality of the uncertainty flagging: does it reflect real model confidence or is it cosmetic
- Whether the product experience feels designed or assembled: the interview will probe the product decisions you made
- The completeness of the output editing and export flow
- Whether you can articulate why this problem requires multiple modalities and what is lost if you remove one

WHAT YOU CANNOT USE

Do not use the following

Pre-built document processing pipelines or OCR services as black boxes. You must understand and control what happens to each modality.

Products that are just wrappers around a single API call with a form in front of them. The pipeline must have real structure.

You may use any multimodal model API (GPT-4o, Claude, Gemini), any frontend framework, and any standard backend tools.

WHAT YOU MUST SUBMIT

- A deployed, working URL we can use right now with sample inputs provided
- GitHub repository with clean commit history
- A 3-minute demo video showing the full product flow including an uncertainty flag and an output edit

- A README that explains the user problem you chose and why it requires multiple modalities, your pipeline architecture and each step's role, how you handle uncertainty, and what you would build next

What to expect in the interview

We will use your product live with inputs you have not seen before. We will ask you to explain why combining the modalities produces better output than processing them separately. We will ask you to describe a case where your pipeline would fail and how you would detect it. We will ask what the product becomes if you remove the AI layer entirely.

TIMELINE

48 hours from the time you receive this document.